# Week 06 Reading: Nonstationary flood frequency at Brays Bayou (bring two questions Wednesday)

### What changes in a flood frequency analysis when the watershed changes

2026-09-28

Read the whole paper before Wednesday's class (9/30), and bring two questions to class: one factual question about something in the paper you did not follow, and one bigger-picture question about what the analysis means or where it could go wrong.

## Assigned

1. Smith et al. (2026) (open access)

You should be able to follow nearly all steps of this paper. One thing to note is the authors fit a log-Pearson Type III (LPIII) distribution as mandated by US federal flood frequency guidelines (Bulletin 17C) require. For this reading, treat it as a lot like the GEV: a three-parameter distribution with a location, a scale, and a shape parameter that controls how heavy the upper tail is, fitted here to the logarithm of the annual peak flows.

The Bayesian estimation, the posterior predictive frequency curves, and the credible intervals are the same ideas you used in labs 4 and 5.

## Questions to bring

1. **Factual.** A question about something specific in the paper: a method, a figure, a number, or a modeling choice you could not follow.
2. **Bigger picture.** A question about what the results mean, whether you would trust them for a decision, or what the analysis leaves out.

## Further reading

1. Section 12.2 of Helsel et al. (2020) (pp. 332-336) for the Mann-Kendall test and regression on time, and section 12.9 (pp. 359-362) for the Red River at Grand Forks, where a strong trend over 1940-2014 looks like one limb of an oscillation over 1882-2014.
2. Section 3.3 of Ghil et al. (2011) for covariates in extreme value parameters.
3. Wong (2018) for storm surge at Sewells Point (Norfolk, Virginia) with several candidate covariates (time, sea level, global mean temperature, and the North Atlantic Oscillation) combined by Bayesian model averaging.

## Bibliography

Ghil, M., Yiou, P., Hallegatte, S., Malamud, B. D., Naveau, P., Soloviev, A., et al. (2011). Extreme Events: Dynamics, Statistics and Prediction. *Nonlinear Processes in Geophysics*, *18*(3), 295–350. https://doi.org/10/fvzxvv

Helsel, D. R., Hirsch, R. M., Ryberg, K. R., Archfield, S. A., & Gilroy, E. J. (2020). *Statistical Methods in Water Resources*. U.S. Geological Survey. https://doi.org/10.3133/tm4A3

Smith, C. H., Skahill, B., & Margo, D. A. (2026). Nonstationary Flood Frequency Analysis for Urban Watersheds Using Open-Source Bayesian Software: Contrasting Case Studies from Texas. *Water*, *18*(5), 636. https://doi.org/10.3390/w18050636

Wong, T. E. (2018). An Integration and Assessment of Multiple Covariates of Nonstationary Storm Surge Statistical Behavior by Bayesian Model Averaging. *Advances in Statistical Climatology, Meteorology and Oceanography*, *4*(1/2), 53–63. https://doi.org/10.5194/ascmo-4-53-2018